Incremental counting method for generating prime numbers

In this context, an *incremental counting method* refers to the ‘sequential traversal of all natural numbers, starting with the number 2’.

The method presented here has the charm of not using multiplication, division or modulo calculation while endlessly generating prime numbers.

For easier reading, digits will be enclosed in quotation marks below, while numbers will be written without quotation marks: ‘2’ means the digit two and 2 means two as number.

Special number system

The *incremental counting method for generating prime numbers* is *based on a special number system*, referred to below as ‘prime number system’:
It is not a place-value system, such as the decimal system, but a special number system in which each position is based on exactly one of the prime numbers – with its own digit pool.

The position based on the prime number 2 consists of the digits ‘0’ and ‘1’. The position based on the prime number 3 consists of the digits ‘0’, ‘1’ and ‘2’.
Or, in general: The position based on the prime number p consists of the digits ‘0’ through ‘p-1’.

Counting in prime number system

Counting in prime number system follows these rules:

- The first number in prime number system is 2 with the digit set consisting of ‘0’ and ‘1’.
- When new places are added to the numbering system, they are initially assigned the digit ‘0’.
- Unlike the place-value system, when incrementing a number in this system, *always all* places are incremented by 1. If the set of digits is exhausted when incrementing a place, the sequence starts again with the digit ‘0’, just as it does in place-value system as well.
- The places in this number system are displaying the remainder of a division of the corresponding number by its base that is the prime number assigned to this position. If none of the digits contain a ‘0’ after incrementing a number, the number is a prime number (because none of the remainder of a division by its base is 0).
  For each prime number found this way, a new place is added (with the digit ‘0’), using the set of digits ‘0’ through ‘p-1’ specific to that prime number.

Start of the count in the example

To display a number in prime number system in readable format in decimal system, two increments are necessary at the same time: One in the prime number system and the other in decimal system.

The table below lists the first 10 numbers in the prime number system.
In the left column, you can see the number in the decimal system. If one number is divisible by a prime number, this is indicated by a (red) digit ‘0’ in a row.
New places in this number system start with the digit ‘0’ in a new column (shown in green).

	Prime number assigned to the position in this number system				
Decimal system	11	7	5	3	2
2					0
3				0	1
4				1	0
5			0	2	1
6			1	0	0
7		0	2	1	1
8		1	3	2	0
9		2	4	0	1
10		3	0	1	0
11	0	4	1	2	1

The first number in prime number system is 2 and is displayed as ‘0’.

The number 3 is obtained by incrementing the number 0 and is represented as ‘1’.
The digit in the only place present is ‘1’, which is not equal to ‘0’. Consequently, 3 is a prime number.
A new place with the digit ‘0’ for the prime number 3 is added.

The number 4 is obtained by incrementing the two places with the digits ‘0’ and ‘1’ and is therefore written as a sequence of digits ‘1’ and ‘0’.
The last place with the digit ‘0’ indicates that the number 4 is divisible by 2.

The number 5 is obtained by incrementing the two places with the digits ‘1’ and ‘0’ and is therefore written as a sequence of digits ‘2’ and ‘1’. Both digits are not equal to ‘0’. Consequently, 5 is a prime number. A new place with the digit ‘0’ for the prime number 5 is added.

And so on.